

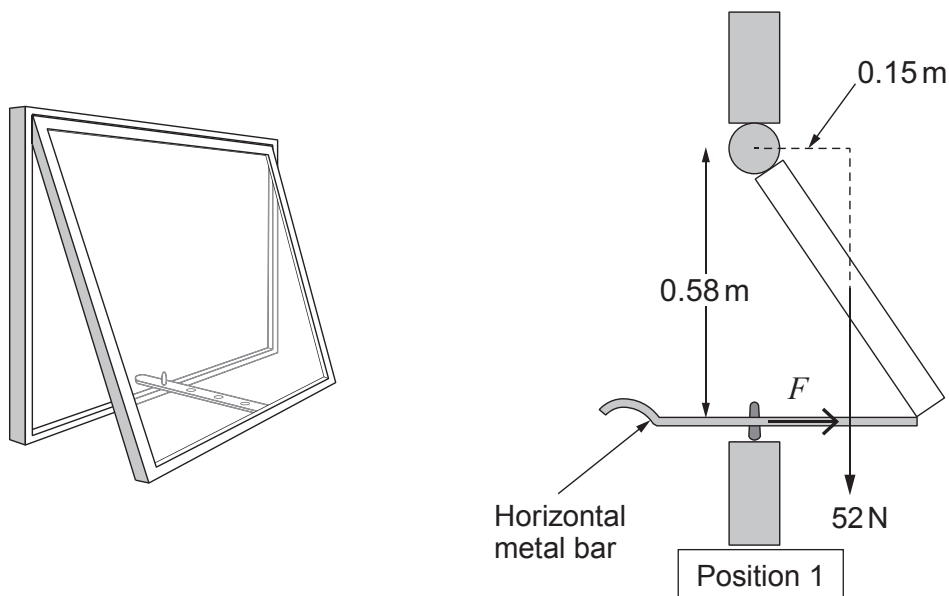
Answer all questions.

1. (a) State in words the equation used to calculate the moment of a force about a point. [1]

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- (b) The picture and diagram show a window hinged at the upper surface. The window is opened by pushing on the horizontal metal bar attached to its lower surface. Holes are drilled into the metal bar so that the window can be supported at various opening positions, one of which is shown below and labelled as Position 1. The hinge provides no resistance to the movement of the window.



- (i) Show that the clockwise moment produced by the weight of the window is approximately 8 Nm . [1]

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- (ii) Hence calculate the force, F , the metal bar exerts on the window. [2]

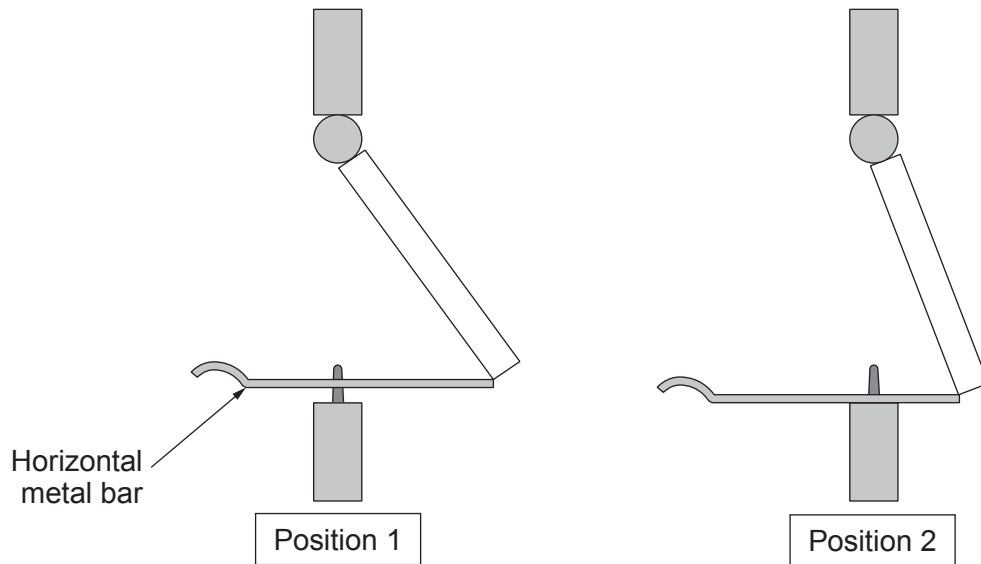
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- (c) Tom and Bethan discuss how the force in the metal bar changes with changing positions. Tom thinks that the force in the bar is greater when the window is in Position 2, whereas Bethan believes that the force is greater when the window is in Position 1. Discuss who is correct, giving a detailed explanation in terms of moments. Assume the metal bar is horizontal in both positions. [4]



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